

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

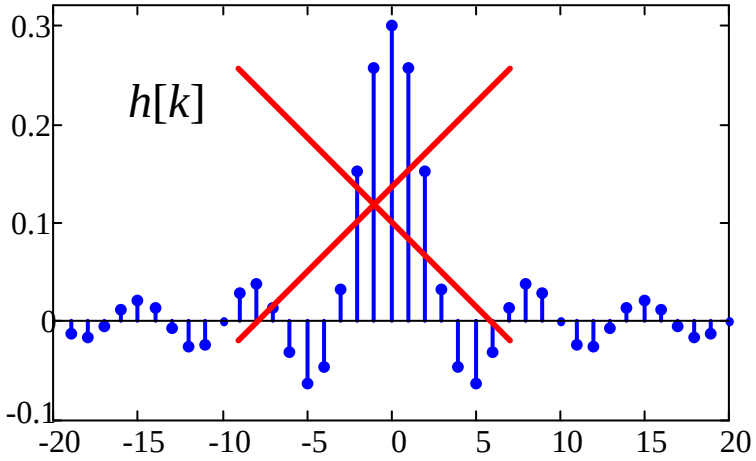
Zhou Hang

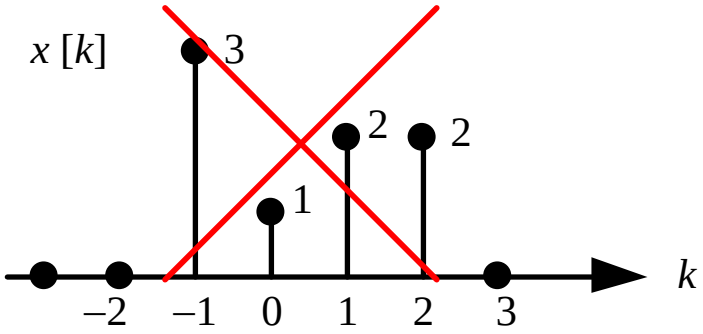
- Why introduce bilateral *z*-transform
- Bilateral *z*-transform and its convergence domain
- The main properties of bilateral *z*-transform
- Calculation of Bilateral inverse *z*-transform

Why introduce bilateral z-transform?

Unilateral *z*-transform $$X(z)=\sum_{k=0}^{\infty}x[k]z^{-k}$$

Limitations: ※ only describe causal systems ※ only analyze one-sided sequences





Bilateral Z-transform and its convergence domain

To more effectively describe discrete LTI systems, a sequence *z*-transform is introduced.

Positive *z*-transformation:

$$X(z) = \sum_{k=-\infty}^{\infty} x[k]\, z^{-k}$$

Inverse *z*-transformation:

$$x[k] = \frac{1}{2\pi \mathrm{j}} \oint_C X(z) z^{k-1} \mathrm{d}z$$

C is a closed curve in the convergence domain of $X(z)$

Bilateral Z-transform and its convergence domain

$$X(z) = \sum_{k=-\infty}^{\infty} x[k]\, z^{-k}$$

The region of *z*-values that can make the above equation converge is called the convergence domain of the *z*-transform,

Abbreviated as ROC (Region of Convergence).

(ROC): $R_- < |z| < R_+$

Bilateral Z-transform and its convergence domain

(1) Finite length sequence

$$X(z)=\sum_{k=N_1}^{N_2}x[k]\,z^{-k} \qquad \mathrm{ROC}\quad 0<|z|<\infty$$

Example: Trying to find the bilateral *z*-transform and convergence domain of rectangular pulse sequence $R_N[k]$

Solution:

$$X(z)=\sum_{k=-\infty}^{+\infty}R_N[k]\,z^{-k}=\sum_{k=0}^{N-1}1\cdot z^{-k}=\frac{1-z^{-N}}{1-z^{-1}},\ |z|>0$$

The convergence domain ROC of the *z*-transform for finite length sequences is generally $0<|z|<+\infty$

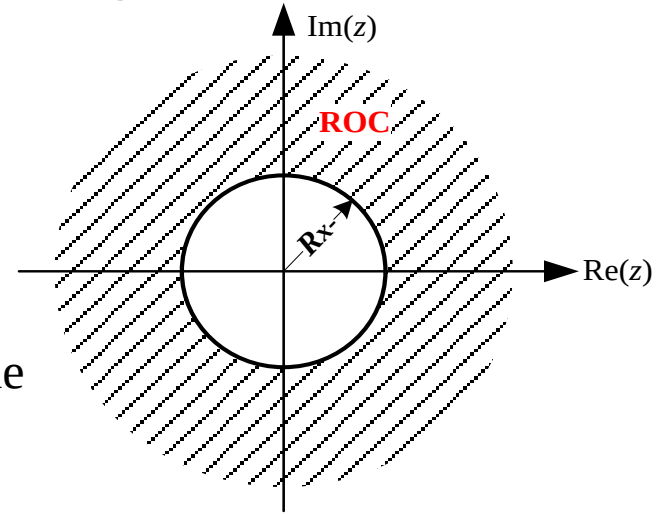
(2) Right-sided sequence

$$X(z)=\sum_{k=N_1}^{\infty} x[k]\,z^{-k}$$

E.g: Attempt to find the bilateral *z*-transform and its convergence domain for exponential sequences $a^k u[k]$

Solution:

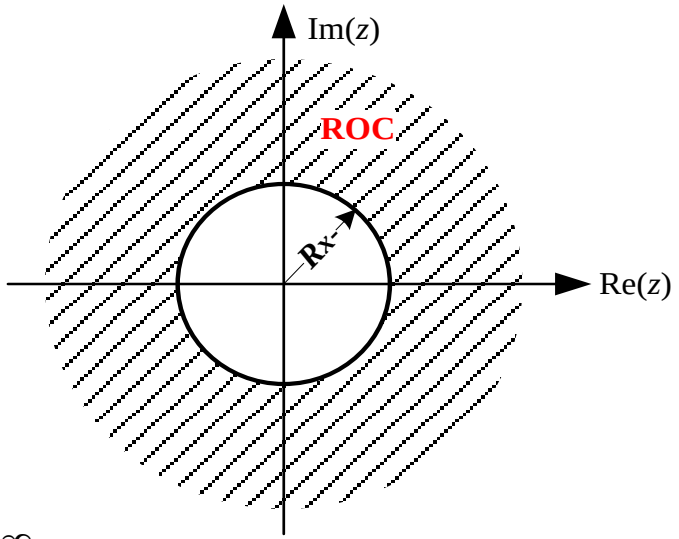
$$X(z)=\sum_{k=0}^{\infty} a^k z^{-k}=\frac{1}{1-az^{-1}} \qquad |z|>|a|$$



The convergence domain is the outer region of a circle with $R_{x-}=|a|$ as its radius.

(2) Right-sided sequence

$$X(z)=\sum_{k=N_1}^{\infty} x[k]\,z^{-k}$$



When $N_1 \geq 0$, the z-transform of the right sequence

The convergence domain is $|z| > R_{x-}$

When $N_1 < 0$, $X(z)$ can be expressed as

$$X(z)=x[N_1]z^{-N_1}+\cdots+x[-1]z+\sum_{k=0}^{\infty} x[k]\,z^{-k}$$

$|z| \neq \infty$, therefore, the convergence domain of the z-transform on the right sequence is $R_{x-} < |z| < \infty$

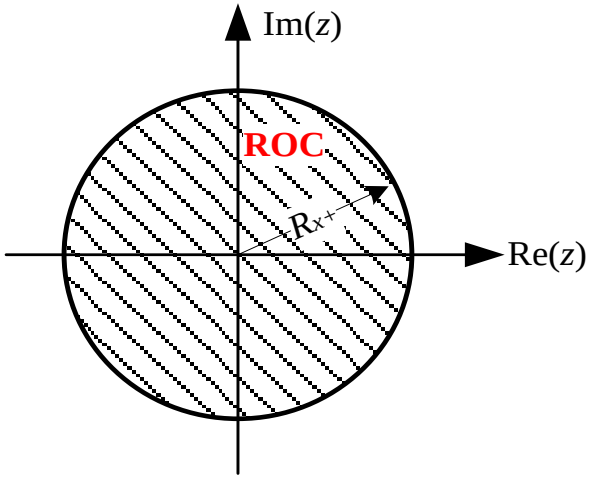
(3) Left-sided sequence $X(z)=\sum_{k=-\infty}^{N_2} x[k]\,z^{-k}$

E.g : Attempt to find the bilateral z-transform of $-b^k\,u[-k-1]$ and its convergence domain

Solution:

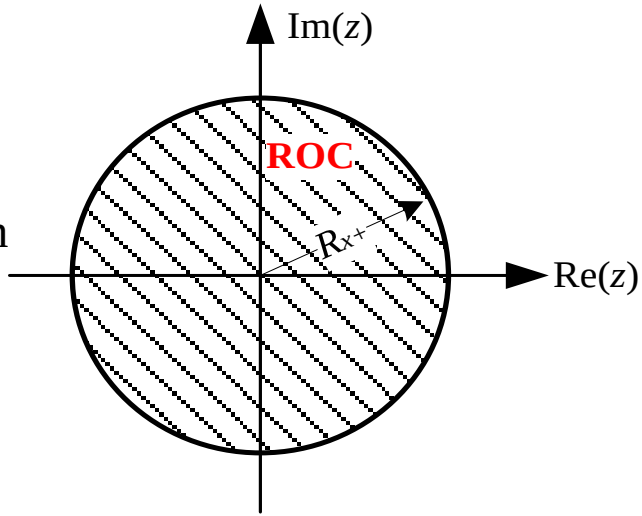
$$X(z)=\sum_{k=-\infty}^{-1} -b^k z^{-k} = \frac{1}{1-bz^{-1}} \quad |z|<|b|$$

The convergence domain is the area within a circle with radius $R_{x+}=|b|$



(3) Left-sided sequence

$$X(z)=\sum_{k=-\infty}^{N_2} x[k]\,z^{-k}$$



When $N_2 \le 0$, the left sequence z undergoes a transformation

The convergence domain is $|z| < R_{x+}$

When $N_2 > 0$, $X(z)$ can be expressed as

$$X(z)=\sum_{k=-\infty}^{-1} x[k]\,z^{-k} + x[0] + x[1]z^{-1} + \cdots + x[N_2]z^{-N_2}$$

$|z| \neq 0$, Therefore, the convergence domain of the z-transform in the left sequence is $0 < |z| < R_{x+}$

(4) Bilateral sequence

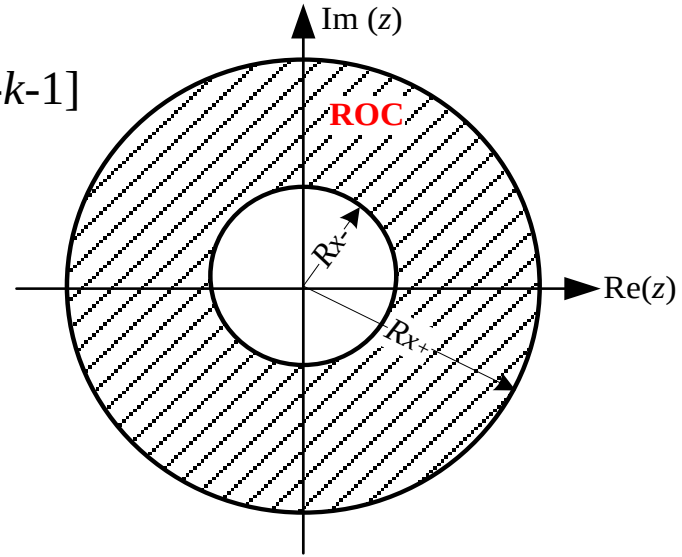
$$X(z)=\sum_{k=-\infty}^{\infty} x[k]\, z^{-k}$$

E.g: Try to find the bilateral z-transform of $a^k\, u[k] - b^k\, u[-k-1]$

Solution: $$X(z)=\frac{1}{1-az^{-1}}+\frac{1}{1-bz^{-1}}$$

The convergence domain of bilateral z-transform is the circular ring region

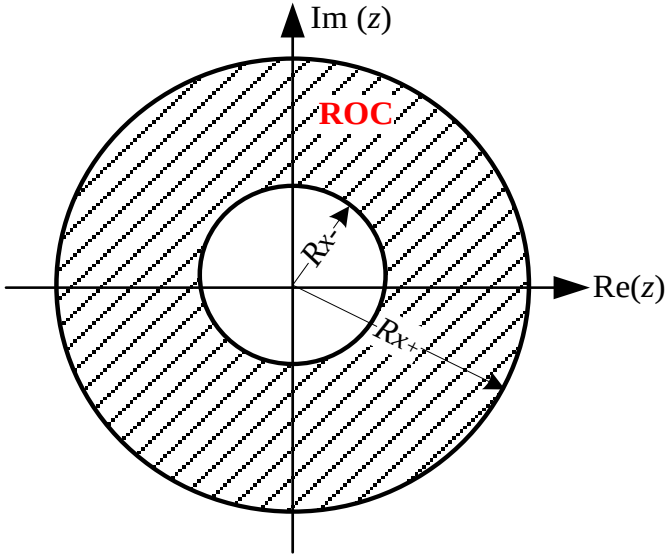
$$|a|<|z|<|b| \qquad (|a|<|b|)$$



(4) Bilateral sequence

$$X(z) = \sum_{k=-\infty}^{\infty} x[k]\, z^{-k}$$

The convergence domain of the bilateral sequence z-transform is: $R_{x-} < |z| < R_{x+}$



The main properties of bilateral z-transform

$$X(z)=\mathscr{Z}\{x[k]\},R_x=\{z;R_{x-}<|z|<R_{x+}\} \qquad Y(z)=\mathscr{Z}\{y[k]\},R_y=\{z;R_{y-}<|z|<R_{y+}\}$$

properties	sequence	Z transform	ROC
linear	$ax[k]+by[k]$	$aX(z)+bY(z)$	$R_x\cap R_y$
adjoint	$x^*[k]$	$X^*(z^*)$	R_x
invert	$x[-k]$	$X(1/z)$	$1/R_{x+}<\|z\|<1/R_{x-}$
shift	$x[k-n]$	$z^{-n}X(z)$	R_x, except $z=0$ or $z=\infty$
convolution	$x[k]*y[k]$	$X(z)Y(z)$	$R_x\cap R_y$
exponential	$a^kx[k]$	$X(z/a)$	$\|a\|R_x$

➢ Displacement characteristics

If $X(z)=\mathcal{Z}\{x[k]\},\quad R_x=\{z;R_{x-}<|z|<R_{x+}\}$

Then $x[k-n]\overset{\mathcal{Z}}{\longleftrightarrow}z^{-n}X(z)\qquad \mathrm{ROC}=R_x$

E.g: Try to find the *z*-transform of the autocorrelation function $r_x[n]$ of sequence $x[k]$.

$$r_x[n]=\sum_k x[k]x[n+k]$$

Solution: The autocorrelation function of sequence $x[k]$ is

Perform *z*-transform on it, there are

$$\mathcal{Z}\{r_x[n]\}=\sum_k x[k]\mathcal{Z}\{x[n+k]\}$$

By utilizing the time-domain displacement properties of the bilateral z-transform, it can be concluded that

$$\mathcal{Z}\{r_x[n]\}=\sum_k x[k]X(z)z^k=X(z^{-1})X(z)$$

The main properties of bilateral z-transform

E.g: Try to find the z-transform of the sequence $x[k]=u[k]-u[k-3]$

Solution:

The z transformation of sequence $u[k]$ is

$$\mathcal{Z}\{u[k]\}=\frac{1}{1-z^{-1}},\quad |z|>1$$

By utilizing the time-domain displacement properties of the bilateral *z*-transform, it can be concluded that

$$\mathcal{Z}\{u[k-3]\}=\frac{z^{-3}}{1-z^{-1}},\quad |z|>1$$

By utilizing the linear properties of the bilateral *z*-transform, it can be concluded that

$$\mathcal{Z}\{x[k]\}=\frac{1-z^{-3}}{1-z^{-1}}\qquad |z|>0$$

The main properties of bilateral z-transform

➢time reversal

If $X(z)=\mathscr{Z}\{x[k]\},\quad R_x=\{z;R_{x-}<|z|<R_{x+}\}$

then $x[-k]\overset{\mathscr{Z}}{\longleftrightarrow}X(z^{-1})\qquad \frac{1}{R_{x+}}<|z|<\frac{1}{R_{x-}}$

E.g: Calculate the *z*-transform of the left sequence $b^k u[-k-1]$.

Solution:

Shift — **time reversal**

$$b^{-k}u[k] \rightarrow b^{-k}u[k-1] \rightarrow b^{k}u[-k-1]$$

$$\mathcal{Z}\{b^{-k}u[k]\} = \sum_{n=0}^{\infty} b^{-k}z^{-k}$$

$$= \frac{1}{1-b^{-1}z^{-1}} \qquad |z| > |b^{-1}|$$

E.g: Calculate the *z*-transform of the left sequence $b^k u[-k-1]$.

Solution: $b^{-k}u[k] \overset{\mathcal{Z}}{\longleftrightarrow} \dfrac{1}{1-b^{-1}z^{-1}} \quad |z| > |b^{-1}|$

By utilizing displacement characteristics, there are

$$b^{-k}u[k-1] = b^{-1}b^{-(k-1)}u[k-1] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{b^{-1}z^{-1}}{1-b^{-1}z^{-1}} \quad |z| > |b^{-1}|$$

By reusing the time reversal feature, we can obtain

$$b^{k}u[-k-1] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{b^{-1}z}{1-b^{-1}z} = -\frac{z}{z-b} = -\frac{1}{1-bz^{-1}} \quad |z| < |b|$$

Calculation of Bilateral Z-Inverse Transform

$$x[k] = \frac{1}{2\pi j}\oint_C X(z)z^{k-1}dz$$

A closed curve in the ROC where C is $X(z)$

※ Residue method

The residue method is relatively complex to calculate, but it has a wide range of applications.

※ Partial fraction expansion method

The partial fraction method is relatively simple to solve, but it is generally only applicable to rational fractions.

Calculation of Bilateral Z-Inverse Transform

※ Partial fraction expansion method for finding the inverse transformation of z

① The sequence z-transformation is decomposed into the sum of partial fractions

② Solve the inverse z-transformation corresponding to each fractional part

$$a^k[k] \overset{\mathscr{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \qquad |z| > |a|$$

$$-b^k[-k-1] \overset{\mathscr{Z}}{\longleftrightarrow} \frac{1}{1-bz^{-1}} = \frac{z}{z-b}, \qquad |z| < |b|$$

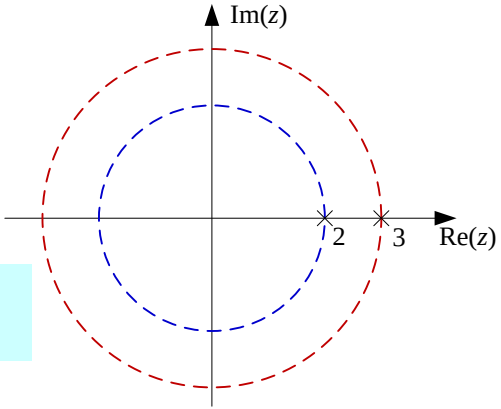
[E.g] Given $X(z)=\dfrac{z^2}{(z-2)(z-3)}$, find $x[k]$ corresponding to different convergence domains.

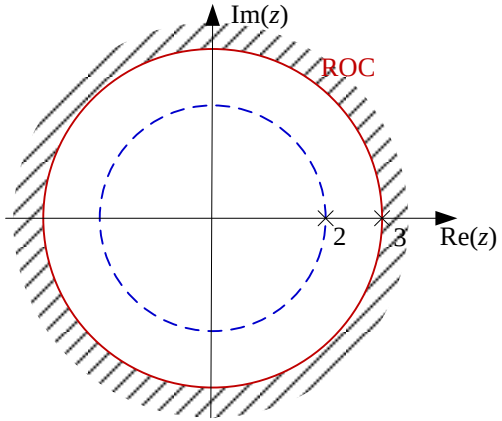
Solution:

$$X(z)=\frac{-2z}{z-2}+\frac{3z}{z-3}$$

$X_1(z)$, pole z=2

$X_2(z)$, pole z=3





(1) $|z|>3$, $X_1(z)$ and $X_2(z)$ both correspond to the sequence on the right

$$x[k]=-2\cdot 2^k u[k]+3\cdot 3^k u[k]$$

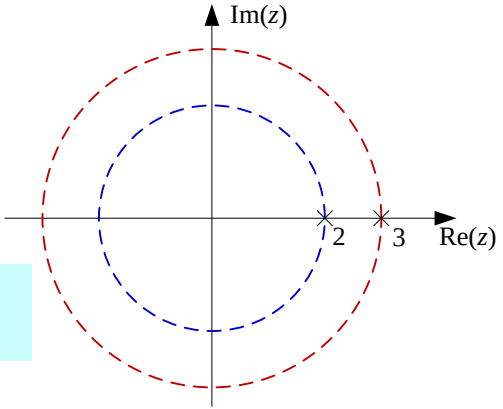
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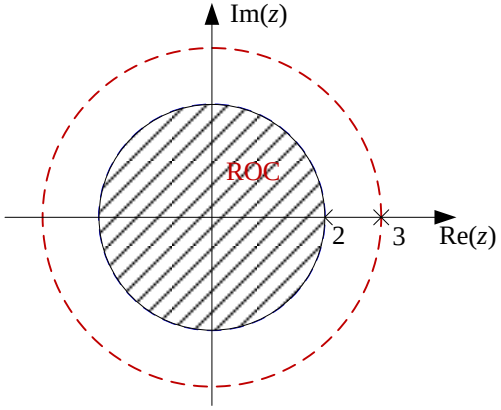
解:

$$X(z)=\frac{-2z}{z-2}+\frac{3z}{z-3}$$

$X_1(z)$, pole z=2

$X_2(z)$, pole z=3





(2) $|z|<2$, $X_1(z)$ and $X_2(z)$ both correspond to the sequence on the left

$$x[k]=2\cdot 2^k u[-k-1]-3\cdot 3^k u[-k-1]$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.